

AI ACADEMIC MENTOR

Course	CSE 4104 — Software Development III
Team	CSE4104-7C-T05
Project Title	AI Academic Mentor
Instructor	Md. Riaz Mahmud, Assistant Professor, Dept. of CSE, NUBTK
Section	7C
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GitHub Repository Link	cse4104-7c-t05-ai-academic-mentor/documentation at main · Priyo-1158/cse4104-7c-t05-ai-academic-mentor · GitHub

Team Information

Role	Name	Student ID	Contribution
Team Leader	Shadman Sadequeen Priyo	11230121158	Project coordination, GitHub management, integration testing
Frontend Developer	Musabbir Hossain Chayon	11230121168	API integration, Report, UI pages, Netlify deployment, Next.js
Backend Developer	SK. Nadirul Haque Nadir	11230121155	Node.js, Express, routing, Express routes, MongoDB models, JWT authentication
AI Integration Lead	Shafin Kabir	11230121175	Gemini API, HuggingFace, AI routes

IMPLEMENTED API ENDPOINTS

Authentication APIs — /api/auth

Method	Endpoint	Auth	Description
POST	/api/auth/register	Public	Register new user account. Returns JWT + user object.
POST	/api/auth/login	Public	Login with email/password. Returns JWT + user object.
GET	/api/auth/me	JWT	Get current authenticated user profile.
PUT	/api/auth/profile	JWT	Update name, studentId, department.
POST	/api/auth/logout	JWT	Stateless logout — instructs client to delete token.

AI Services APIs — /api/ai

Method	Endpoint	Auth	Description
POST	/api/ai/chat	Public	AI academic chat. Gemini → HuggingFace → Smart Fallback chain.
POST	/api/ai/quiz	Public	Generate AI quiz on given topic.
POST	/api/ai/summarize	Public	Summarize provided text using AI.
POST	/api/ai/plan	Public	Generate a study plan for given subject.

Quiz Results APIs — /api/quiz

Method	Endpoint	Auth	Description
GET	/api/quiz	JWT	List all quiz results for authenticated user.
POST	/api/quiz	JWT	Save a quiz result. Req: topic, score, numQuestions.
GET	/api/quiz/:id	JWT	Get single quiz result by ID.
DEL	/api/quiz/:id	JWT	Delete a quiz result.

Study Plan APIs — /api/plan

Method	Endpoint	Auth	Description
GET	/api/plan	JWT	List all study plans for user.
POST	/api/plan	JWT	Create/save a study plan.
GET	/api/plan/:id	JWT	Get single study plan.
PUT	/api/plan/:id	JWT	Update a study plan.
DEL	/api/plan/:id	JWT	Delete a study plan.

Notes / Summary APIs — /api/notes & /api/summary

Method	Endpoint	Auth	Description
GET	/api/notes	JWT	List all user notes.
POST	/api/notes	JWT	Create note. Req: title, originalText.
GET	/api/notes/:id	JWT	Get single note.
PUT	/api/notes/:id	JWT	Update a note.
DEL	/api/notes/:id	JWT	Delete a note.
GET	/api/summary	JWT	List summaries (excludes originalText for performance).
POST	/api/summary	JWT	Save a summary. Req: originalText.
GET	/api/summary/:id	JWT	Get single summary.
DEL	/api/summary/:id	JWT	Delete a summary.

Progress APIs — /api/progress

Method	Endpoint	Auth	Description
GET	/api/progress	JWT	Aggregated progress: quiz count, avg score, plan count, notes count.
GET	/api/progress/stats	JWT	Last 10 quiz scores with topic and date.

Admin APIs — /api/admin (Admin Role Required)

Method	Endpoint	Auth	Description
GET	/api/admin/stats	Admin	Platform stats: total users, quizzes, plans, notes, avg score.
GET	/api/admin/users	Admin	Paginated user list. Query: ?page=1&limit=20&role=student.
GET	/api/admin/users/:id	Admin	Get single user detail.
PUT	/api/admin/users/:id	Admin	Update user role and/or isActive status.
DEL	/api/admin/users/:id	Admin	Delete user. Cannot delete own account.

Health Check

Method	Endpoint	Auth	Description
GET	/api/health	Public	Returns API status, DB connection state, version info, and all endpoint paths.